

# ENCRYPTODERA FINANCIAL

## Cyber Incident Investigation Report

Confidential | Internal Use Only

Investigation Period: January 2024 – February 2024

### 1. Executive Summary

This report documents findings from a cybersecurity investigation at Encryptodera Financial. Three distinct threat campaigns were identified and investigated using KQL across multiple log sources including authentication events, email logs, process events, network flows, and file creation events.

| Case                  | Threat Actor            | Attack Type                 | Status    |
|-----------------------|-------------------------|-----------------------------|-----------|
| 1 – Offensive Odor    | Barry Shmelly (Insider) | Data Exfiltration           | Confirmed |
| 2 – Crypto Conquest   | External Threat Actor   | Ransomware                  | Confirmed |
| 3 – A Network Mystery | Jane Smith (Insider)    | Crypto Theft / Exfiltration | Confirmed |

### 2. Case 1: Offensive Odor – Insider Data Theft

#### 2.1 Threat Actor Profile

| Field      | Detail                                  |
|------------|-----------------------------------------|
| Name       | Barry Shmelly                           |
| Email      | barry_shmelly@encryptoderafinancial.com |
| Username   | bashmelly                               |
| IP Address | 10.10.0.1                               |
| Hostname   | IGOY-DESKTOP                            |
| Role       | StackOverflow Copy Paster               |

#### 2.2 Timeline of Events

| Date         | Activity                                                                                        |
|--------------|-------------------------------------------------------------------------------------------------|
| Dec 26, 2023 | Researched file transfer methods on cybersecurity-insiders.com                                  |
| Jan 15, 2024 | Downloaded 7-Zip installer (7z2002-x64.exe) and researched USB flash drives via WikiHow         |
| Jan 15, 2024 | Downloaded SECRET_MergersAndAcquisitions_Strategy2025.docx and ExecutiveSalaryNegotiations.docx |
| Jan 15, 2024 | Downloaded Encryptodera_Proprietary_Algorithms.zip                                              |
| Jan 16, 2024 | Sent phishing/suspicious email to software engineering team                                     |
| Jan 18, 2024 | Sent angry email (subject: YOU ARE A GREEDY PIG...) to Les Goh (CFO)                            |
| Jan 18, 2024 | Sent malicious Company_Financials_Q1_2024_Review.xlsx.exe to 9 recipients                       |
| Feb 01, 2024 | Barry's account accessed from external IP 143.38.175.105                                        |

## 2.3 Key Findings

- Sensitive documents compressed and password-protected using 7-Zip with password: securePass123
- Files stored on personal removable drive at path: E:\SchmellyDrive\
- Targeted employees: Christopher, Michelle, Jarrod (Software Engineers) and Les Goh (CFO)
- 9 malicious phishing emails sent internally with executable attachments disguised as Office files
- Barry researched data exfiltration methods and USB usage prior to the theft

## 3. Case 2: Crypto Conquest – Ransomware Attack

### 3.1 Attack Overview

An external threat actor gained initial access via a malicious attachment sent by Barry Shmelly, deployed remote access tooling, conducted internal reconnaissance, then deployed ransomware across 306 machines. The ransomware appended the extension .umadbrot to encrypted files and dropped a ransom note: YOU\_GOT\_CRYTOED\_SO\_GIMME\_CRYPT0.txt.

### 3.2 Timeline of Events

| Date / Time (UTC)  | Activity                                                                                     |
|--------------------|----------------------------------------------------------------------------------------------|
| Feb 01, 2024 08:50 | Company_Financials_Q1_2024_Review.xlsx.exe dropped on 41QI-LAPTOP via Barry's phishing email |

| Date / Time (UTC)  | Activity                                                                                          |
|--------------------|---------------------------------------------------------------------------------------------------|
| Feb 01, 2024 08:50 | screenconnect_client.exe appeared seconds later on 41QI-LAPTOP – remote access established        |
| Feb 01, 2024       | Barry's account (bashmelly) accessed from external IP 143.38.175.105 via MAIL-SERVER01            |
| Feb 02, 2024 03:32 | First discovery command: systeminfo executed on 41QI-LAPTOP (8 machines total)                    |
| Feb 02, 2024       | Attacker ran: cmd.exe /C nltest /dclist:encryptoderafinancial.com to locate Domain Controller     |
| Feb 02, 2024       | ScreenConnect deployed on 3 machines: QW91-MACHINE, 41QI-LAPTOP, 3RMK-DESKTOP                     |
| Feb 02, 2024       | lihenry_domain_admin credentials used to access DOMAIN_CONTROLLER_SERVER                          |
| Feb 02, 2024       | Mimikatz executed on GJ95-LAPTOP as: totally_not_mimikatz.exe "sekurlsa::logonpasswords"          |
| Feb 17, 2024 02:29 | Base64 PowerShell executed: downloaded files_go_byebye.exe from notification-finance-services.com |
| Feb 17, 2024 02:30 | files_go_byebye.exe appeared on UL8R-MACHINE                                                      |
| Feb 17, 2024 02:34 | Ransom note first observed. 50 files encrypted on UL8R-MACHINE with .umadbro extension            |
| Feb 17, 2024       | gpupdate /force executed on 306 machines – mass ransomware deployment via Group Policy            |

### 3.3 Lateral Movement & Credential Theft

- Initial access via malicious .xlsx.exe attachment (MITRE T1566.001 - Spearphishing Attachment)
- ScreenConnect used for persistent remote access across 3 machines
- Mimikatz (S0002) used for credential dumping: MITRE T1003.001 - LSASS Memory
- Compromised domain admin account (lihenry\_domain\_admin) used to reach Domain Controller
- systadmi\_local\_admin credentials abused by non-IT employee Robin Kirby (Risk Analyst)
- 15 days elapsed between first discovery command (Feb 2) and ransomware deployment (Feb 17)
- Valerie Orozco (GJ95-LAPTOP) was phished but did NOT click the malicious link

## 4. Case 3: A Network Mystery – Cryptocurrency Theft

## 4.1 Threat Actor Profile

| Field                | Detail                              |
|----------------------|-------------------------------------|
| Name                 | Jane Smith                          |
| Username             | jasmith                             |
| IP Address           | 10.10.0.2                           |
| Hostname             | GOTI-LAPTOP                         |
| Role                 | Blockchain Contractor (Crypto Bruh) |
| Accomplice           | elboss@westealurcrypto.com          |
| Exfil Destination IP | 182.56.23.121 (port 21 / FTP)       |

## 4.2 Timeline of Events

| Date           | Activity                                                                        |
|----------------|---------------------------------------------------------------------------------|
| Jan 21, 2024   | First data sent to external IP 182.56.23.121 via FTP (port 21)                  |
| Multiple dates | Data exfiltrated across 27 distinct days to 182.56.23.121 (208,138 total bytes) |
| Investigation  | Jane searched for company cold storage crypto wallet locations                  |
| Investigation  | Jane in email contact with elboss@westealurcrypto.com who provided the exfil IP |
| Investigation  | Jane downloaded ftp_client.exe and crypto_stealer.exe                           |
| Feb 05, 2024   | Largest single-day data transfer: 12,716 bytes to 182.56.23.121                 |
| Final stage    | crypto_stealer.exe executed daily to steal crypto wallet data                   |

## 4.3 Exfiltration Details

- Data staged locally at: C:\Users\jasmith\ToTheMoon\
- Source data path:  
\\NetworkShare\critical\_infrastructure\Crypto\_Wallet\_Storage\_Locations\
- Tool executed with --daily flag for scheduled recurring exfiltration
- Total data sent to external IP: 208,138 bytes over 27 days
- Only one internal IP (10.10.0.2 / Jane) communicated with the exfil server
- FTP (port 21) used – unencrypted protocol, credentials transmitted in plaintext

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## 5. Recommendations

## 5.1 Immediate Actions

- Terminate and revoke all credentials for Barry Shmelly and Jane Smith
- Isolate and reimage all 306 machines affected by ransomware
- Block external IP 182.56.23.121 and domain notification-finance-services.com at firewall
- Rotate all domain admin credentials, especially lihenry\_domain\_admin
- Audit all accounts that used systadmi\_local\_admin credentials

## 5.2 Long-Term Improvements

- Implement DLP (Data Loss Prevention) to flag bulk file downloads and USB usage
- Block or alert on outbound FTP (port 21) traffic to external IPs
- Deploy EDR solution to detect Mimikatz and credential dumping attempts
- Enforce MFA on all accounts, especially domain admin accounts
- Monitor for encoded PowerShell execution (-enc flag) across all endpoints
- Conduct regular phishing awareness training for all staff
- Restrict Group Policy modification rights to prevent mass ransomware deployment

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## 6. Log Sources Used

| Table                 | Purpose                                                           |
|-----------------------|-------------------------------------------------------------------|
| AuthenticationEvents  | Login attempts, credential abuse, lateral movement                |
| Email                 | Phishing emails, suspicious communications, malicious attachments |
| ProcessEvents         | Command execution, Mimikatz, encoded PowerShell, ransomware       |
| FileCreationEvents    | Ransomware file drops, malicious executables, encrypted files     |
| OutboundNetworkEvents | Browsing activity, C2 beaconing, data staging                     |
| InboundNetworkEvents  | Malicious file downloads                                          |
| NetworkFlow           | FTP exfiltration traffic, bytes transferred                       |
| Employees             | IP-to-user mapping, role correlation                              |
| PassiveDns            | Suspicious domain resolution                                      |